

Note

Eight-Week Creatine-Glucose Supplementation Alleviates Clinical Features of Long COVID

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Summary Preliminary studies demonstrated beneficial effects of dietary creatine across different post-viral fatigue syndromes. Creatine is often co-administered with glucose to improve its potency yet whether glucose boost the efficacy of creatine in long COVID remains currently unknown. In this report, we investigate the effects of 8-wk creatine intake with and without glucose on patient-reported outcomes, exercise tolerance, and tissue creatine levels in patients with long COVID. Fifteen male and female long COVID adult patients (age 39.7 ± 16.0 y; 9 women) with moderate fatigue and at least one of additional long COVID-related symptoms volunteered to participate in this randomized controlled parallel-group interventional trial. All patients were allocated in a double-blind parallel-group design (1 : 1 : 1) to receive creatine (8 g of creatine monohydrate per day), a mixture of creatine and glucose (8 g of creatine monohydrate and 3 g of glucose per day), or placebo (3 g of glucose per day) t.i.d. during an 8-wk intervention interval. Two-way ANOVA with repeated measures (treatment vs. time interaction) revealed significant differences in changes in total creatine levels between the groups, showing an interaction effect at two brain locations (right precentral white matter $F=34.740$, $p=0.008$; partial $\eta^2=0.72$; left paracentral grey matter $F=19.243$, $p=0.019$; partial $\eta^2=0.88$), with creatine and creatine-glucose outcompeted placebo to elevate creatine levels at these two locations. Several long COVID symptoms (including body aches, breathing problems, difficulties concentrating, headache, and general malaise) were significantly reduced in creatine-glucose group at 8-wk follow-up ($p \leq 0.05$); the effect sizes for reducing body aches, difficulties concentrating, and headache were 1.33, 0.80, and 1.12, respectively, suggesting a large effect of creatine-glucose mixture for these outcomes. Our preliminary findings suggest that supplying exogenous creatine with glucose could be recommended as an effective procedure in replenishing brain creatine pool and alleviating long COVID features in this prevalent condition.

Key Words long COVID, creatine, brain, muscle, fatigue, glucose

Long COVID (also known as post-COVID-19 syndrome) is a broad term that describes the effects of COVID-19 that lasts for several weeks or months beyond the initial illness (1). The most commonly reported long COVID symptom continues to be fatigue (72% of patients), followed by difficulty concentrating (51%), muscle ache (49%), and shortness of breath (48%) (2). While no gold standard medication is currently available to treat long COVID, several management options are available to help relieve symptoms of this perplexing and prevalent condition, including dietary interventions (3). Among various nutrients, dietary creatine has been purported as a possible adjunct therapeutics that

can help in post-COVID-19 rehabilitation (4). Creatine is considered a semi-essential nutrient that plays a key role in energy metabolism under many physiological and pathological conditions (5), including post-viral fatigue syndromes. A recent randomized-controlled trial demonstrated beneficial effects of creatine supplementation in long COVID (6), with 4 g of creatine monohydrate per day administered for 6 mo outcompeted placebo to alleviate breathing difficulties, body aches, and difficulties concentrating, and augment creatine levels across the body in patients with the disease. Creatine is often co-administered with glucose to improve its potency in both experimental and clinical nutrition (7, 8). However, whether glucose boosts the efficacy of creatine in long COVID is unclear. In this pilot study, we investigated whether the administration

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of creatine for 8 wk with and without glucose affects patient-reported outcomes, exercise tolerance, and tissue creatine levels differently in patients with long COVID.

Materials and Methods

Fifteen ($n=15$) male and female long COVID adult patients (age 39.7 ± 16.0 y, weight 74.0 ± 9.7 kg, height 173.9 ± 8.8 cm; 9 women) with moderate fatigue and at least one of additional long COVID-related symptoms (e.g., ageusia, anosmia, body aches, breathing difficulties, difficulties concentrating, headache, lung pain, general malaise) volunteered to participate in this randomized controlled parallel-group interventional trial. All patients were allocated in a double-blind parallel-group design (1 : 1 : 1) to receive creatine (experimental group 1; 8 g of creatine monohydrate per day), a mixture of creatine and glucose (experimental group 2; 8 g of creatine monohydrate and 3 g of glucose per day), or glucose (control group; 3 g of glucose per day) during an 8-wk intervention interval. A dosage of creatine (8 g/d) and intervention duration (8 wk) was determined as a risk-free protocol administered across various neurological conditions (9). Creatine monohydrate and glucose were provided by Alzchem Group AG (Creavitalis™, Trostberg, Germany) and Inexall Company LLC (Belgrade, Serbia), respectively, with interventions similar in appearance, texture and sensory characteristics. All participants refrained from using any other dietary supplements or long COVID therapeutics during the study. The main outcomes were the concentrations of total creatine assessed by magnetic resonance spectroscopy (MRS) in the vastus medialis muscle and thalamus, frontal, precentral, paracentral, and parietal white and grey matter of the brain. Fatigue was assessed with Multidimensional Fatigue Inventory (MFI-20) test, and the severity of long COVID symptoms was evaluated with the visual analogue scale (VAS) scores; walking time to exhaustion was measured during an incremental test on a motorized treadmill. The total creatine (creatine plus phosphocreatine) levels were measured using a single-voxel proton MRS at 1.5 T, employing the matrix head coil in circularly polarized mode on the Avanto Scanner (Siemens, Erlangen, Germany). Detailed information about experimental protocols used is available in a primary trial (6); the current study was an extension arm focused to explore alternative creatine intake regimens in long COVID. The study was conducted according to the guidelines laid down in the Declaration of Helsinki, and all procedures involving human participants were approved by the local IRB at the University of Novi Sad (#46-06-03/2022A-2-E2). Data were initially assessed for normality (Shapiro–Wilk test) and variance homogeneity (Bartlett’s test). When homogeneity of variances was confirmed for normally distributed data, interaction effects (time [pre, post] vs. intervention [experimental group 1, experimental group 2, control group]) were assessed through two-way ANOVA with repeated measures. In cases of non-homogeneous variances, data were analyzed using

the Friedman test. Post hoc tests (LSD and Wilcoxon) compared sample pairs. Significance was set at $p \leq 0.05$. A Bonferroni correction was utilized to mitigate the risk of Type I error due to multiple comparisons. Cohen’s statistics determined effect sizes ($d \geq 0.8$ for large effects), with missing data excluded from analyses.

Results

All 15 patients completed the trial. No participants reported any significant adverse events or side effects during the trial. The compliances with the interventional regimens were $93.2 \pm 5.0\%$ (experimental group 1), $88.2 \pm 9.3\%$ (experimental group 2), and $87.6 \pm 8.7\%$ (control group) ($p=0.49$). The changes in study outcomes during the trial are depicted in Table 1. Tissue creatine levels significantly increased at three locations after the study for experimental group 1 ($p \leq 0.05$); the Cohen’s effect size was above the threshold for medium effect in parietal mesial grey matter only ($d=0.65$). Total creatine levels increased significantly at eight locations following the intervention in experimental group 2 as compared to the baseline concentrations ($p \leq 0.05$). The effect sizes varied across locations, with medium-to-large effects of creatine-glucose mixture reported at the right frontal grey matter ($d=0.54$), left precentral white matter ($d=0.66$), right precentral white matter ($d=0.60$), left parietal white matter ($d=0.97$), and right parietal white matter ($d=0.77$). No significant changes in creatine levels were found in control group ($p > 0.05$). Two-way ANOVA with repeated measures (treatment vs. time interaction) revealed significant differences for changes in total creatine levels at right precentral white matter and left paracentral grey matter between the groups for an interaction effect (right precentral white matter $F=34.740$, $p=0.008$; partial $\eta^2=0.72$; left paracentral grey matter $F=19.243$, $p=0.019$; partial $\eta^2=0.88$), with experimental groups 1 and 2 outcompeted control group to elevate creatine levels at these two locations. A non-significantly higher creatine level was also found for other locations in experimental groups as compared to the control group ($p < 0.20$), including the vastus medialis muscle, left precentral white matter, right paracentral grey matter, left parietal white matter, and left and right parietal mesial grey matter. In addition, no differences for an amplitude of changes in creatine levels were found between experimental group 1 and 2 ($p > 0.05$).

The mean time to exhaustion was significantly improved for 20 s (95% CI, from -173 to 213) in experimental group 1 ($p=0.03$). This was accompanied by a significant reduction in fatigue scores for the reduced activity subdomain in this group ($p=0.008$), with $d=1.52$ implying a large effect of creatine for the variable. In addition, body aches, difficulties concentrating, and general malaise were significantly reduced in experimental group 1 at post-administration ($p \leq 0.05$); the Cohen’s effect sizes for all three variables were above a threshold value of 0.80, suggesting a large effect size. Several long COVID symptoms (including body aches, breathing problems, difficulties concentrating, head-

Table 1. Study outcomes during the trial. Values are mean $\pm$ SD.

	Creatine (n=5)		Creatine plus glucose (n=5)		Glucose (n=5)		p*
	Baseline	8-wk	Baseline	8-wk	Baseline	8-wk	
Total creatine (mM)							
Vastus medialis muscle	29.0 $\pm$ 5.6	30.4 $\pm$ 5.2 [†]	30.0 $\pm$ 6.7	32.3 $\pm$ 6.8 [†]	27.8 $\pm$ 4.2	27.0 $\pm$ 4.2	0.14
Thalamus	5.1 $\pm$ 1.0	5.3 $\pm$ 0.9	5.0 $\pm$ 0.8	5.1 $\pm$ 0.8 [†]	5.5 $\pm$ 0.5	5.5 $\pm$ 0.7	0.86
Left frontal white matter	6.7 $\pm$ 0.9	6.8 $\pm$ 0.9	6.9 $\pm$ 0.9	7.2 $\pm$ 0.7	5.8 $\pm$ 0.4	5.6 $\pm$ 1.0	0.68
Right frontal white matter	6.7 $\pm$ 1.0	7.0 $\pm$ 1.1	6.6 $\pm$ 0.8	6.8 $\pm$ 0.9	5.8 $\pm$ 0.8	5.6 $\pm$ 0.8	0.59
Left frontal grey matter	6.6 $\pm$ 0.6	6.5 $\pm$ 0.5	6.6 $\pm$ 0.8	6.8 $\pm$ 0.9	6.8 $\pm$ 0.4	6.3 $\pm$ 0.9	0.33
Right frontal grey matter	6.6 $\pm$ 1.1	6.7 $\pm$ 1.0	6.8 $\pm$ 1.0	7.4 $\pm$ 1.2 [†]	7.0 $\pm$ 0.9	7.0 $\pm$ 0.7	0.22
Left precentral white matter	6.4 $\pm$ 1.0	6.7 $\pm$ 0.8	5.9 $\pm$ 0.5	6.3 $\pm$ 0.7 [†]	5.6 $\pm$ 0.6	5.6 $\pm$ 0.5	0.07
Right precentral white matter	6.8 $\pm$ 0.9	7.4 $\pm$ 1.1	6.2 $\pm$ 1.0	6.8 $\pm$ 1.0 [†]	5.4 $\pm$ 0.6	5.3 $\pm$ 0.7	0.01
Left paracentral grey matter	6.8 $\pm$ 0.8	6.9 $\pm$ 0.9	7.3 $\pm$ 1.1	7.9 $\pm$ 1.4	7.0 $\pm$ 0.9	6.9 $\pm$ 0.7	0.02
Right paracentral grey matter	7.2 $\pm$ 0.8	7.6 $\pm$ 1.6	7.1 $\pm$ 1.3	7.7 $\pm$ 1.4 [†]	6.4 $\pm$ 0.7	6.2 $\pm$ 1.1	0.12
Left parietal white matter	7.5 $\pm$ 1.6	7.8 $\pm$ 1.3	5.2 $\pm$ 0.8	6.3 $\pm$ 1.4 [†]	5.4 $\pm$ 0.6	5.3 $\pm$ 0.5	0.19
Right parietal white matter	7.6 $\pm$ 1.9	7.7 $\pm$ 1.9	5.1 $\pm$ 0.7	5.6 $\pm$ 0.6	5.2 $\pm$ 0.7	5.2 $\pm$ 0.7	0.63
Left parietal mesial grey matter	8.1 $\pm$ 1.3	8.6 $\pm$ 1.1 [†]	8.7 $\pm$ 1.2	9.2 $\pm$ 1.2	7.6 $\pm$ 0.7	7.4 $\pm$ 0.6	0.09
Right parietal mesial grey matter	8.4 $\pm$ 1.4	9.2 $\pm$ 1.0 [†]	8.4 $\pm$ 1.1	8.9 $\pm$ 1.0 [†]	7.3 $\pm$ 0.6	7.0 $\pm$ 1.0	0.07
Time to exhaustion (s)	824 $\pm$ 138	844 $\pm$ 127 [†]	872 $\pm$ 61	899 $\pm$ 58	915 $\pm$ 116	908 $\pm$ 106	0.14
Fatigue (score)							
General fatigue	12.6 $\pm$ 1.1	12.0 $\pm$ 1.6	12.2 $\pm$ 1.5	11.6 $\pm$ 0.9	12.0 $\pm$ 1.2	11.6 $\pm$ 0.9	0.62
Physical fatigue	12.2 $\pm$ 0.8	11.4 $\pm$ 1.1	13.8 $\pm$ 2.2	13.0 $\pm$ 1.0	12.8 $\pm$ 0.8	12.6 $\pm$ 0.5	0.40
Reduced motivation	13.4 $\pm$ 0.5	12.2 $\pm$ 1.5	13.0 $\pm$ 1.9	12.8 $\pm$ 1.6	11.4 $\pm$ 1.8	11.0 $\pm$ 1.0	0.55
Reduced activity	13.6 $\pm$ 1.1	12.0 $\pm$ 1.0 [†]	12.6 $\pm$ 2.3	10.6 $\pm$ 0.9	12.6 $\pm$ 1.1	12.6 $\pm$ 1.5	0.13
Mental fatigue	11.6 $\pm$ 0.5	10.3 $\pm$ 1.3	11.4 $\pm$ 1.8	11.2 $\pm$ 1.8	10.0 $\pm$ 1.4	10.8 $\pm$ 1.3	0.15
Symptoms (points)							
Ageusia	0.0 $\pm$ 0.0	0.2 $\pm$ 0.4	0.2 $\pm$ 0.4	0.0 $\pm$ 0.0	2.2 $\pm$ 2.9	0.6 $\pm$ 0.9	0.28
Anosmia	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.2 $\pm$ 0.4	0.4 $\pm$ 0.5	2.6 $\pm$ 2.7	1.0 $\pm$ 1.2 [†]	0.14
Body aches	2.8 $\pm$ 2.2	0.8 $\pm$ 1.3 [†]	4.0 $\pm$ 1.0	2.8 $\pm$ 0.8 [†]	1.2 $\pm$ 1.3	0.6 $\pm$ 0.9 [†]	0.20
Breathing difficulties	2.6 $\pm$ 3.1	0.4 $\pm$ 0.9	2.0 $\pm$ 2.0	1.4 $\pm$ 1.5 [†]	1.4 $\pm$ 1.1	0.8 $\pm$ 0.8	0.61
Difficulties concentrating	4.2 $\pm$ 1.3	2.2 $\pm$ 1.6 [†]	3.2 $\pm$ 1.9	1.6 $\pm$ 2.1 [†]	4.4 $\pm$ 3.0	3.0 $\pm$ 2.4	0.71
Headache	3.2 $\pm$ 3.5	0.8 $\pm$ 1.1	5.0 $\pm$ 2.5	2.6 $\pm$ 1.7 [†]	3.8 $\pm$ 3.3	2.8 $\pm$ 1.9	0.66
Lung pain	1.8 $\pm$ 2.5	0.6 $\pm$ 0.9	0.4 $\pm$ 0.5	0.0 $\pm$ 0.0	0.4 $\pm$ 0.5	0.6 $\pm$ 0.5	0.20
General malaise	4.2 $\pm$ 2.4	1.0 $\pm$ 0.7 [†]	4.8 $\pm$ 2.3	3.8 $\pm$ 1.9 [†]	4.6 $\pm$ 2.7	3.0 $\pm$ 1.6	0.18

Asterisk (*) indicates statistical significance for interaction effect (time [two assessment points] vs. treatment [three groups]).

Dagger (†) indicates a significant difference at $p \leq 0.05$ for within-group comparisons (baseline vs. post-administration).

ache, and general malaise) were significantly reduced in experimental group 2 at 8 wk follow-up ($p \leq 0.05$); the effect sizes for reducing body aches, difficulties concentrating, and headache were 1.33, 0.80, and 1.12, respectively, suggesting a large effect of creatine-glucose mixture for these outcomes. The participants receiving glucose experienced no significant changes in most patient-reported outcomes at 8 wk follow-up ($p > 0.05$), except for a significant reduction in anosmia ($p = 0.05$) and body aches ($p = 0.04$). Two-way ANOVA with repeated measures (treatment vs. time interaction) revealed no significant differences for time to exhaustion, fatigue subdomains, and patient-reported symptoms between three groups ($p > 0.05$), yet the participants from both experimental groups tended to be superior to the participants from control group to extend the time to exhaustion; reduce two fatigue subdomains (reduced activity and mental fatigue); and attenuate

body aches, lung pain, and general malaise ($p \leq 0.20$). The effect sizes in two experimental groups for all outcomes evaluated during the trial are depicted in Fig. 1.

Discussion

The most important findings of this trial was that medium-term intake of dietary creatine with or without glucose was a well-tolerated intervention that increases tissue creatine levels in patients suffering from long COVID. The effects of creatine (sole or co-administered with glucose) were superior compared to the control intervention for increasing creatine levels at several brain locations that includes both white and grey matter. Since long COVID is characterized by reduced tissue creatine (10), supplying exogenous creatine could be recommended as an effective procedure in replenishing brain creatine pool in this disorder. Although no significant differences were reported between creatine and

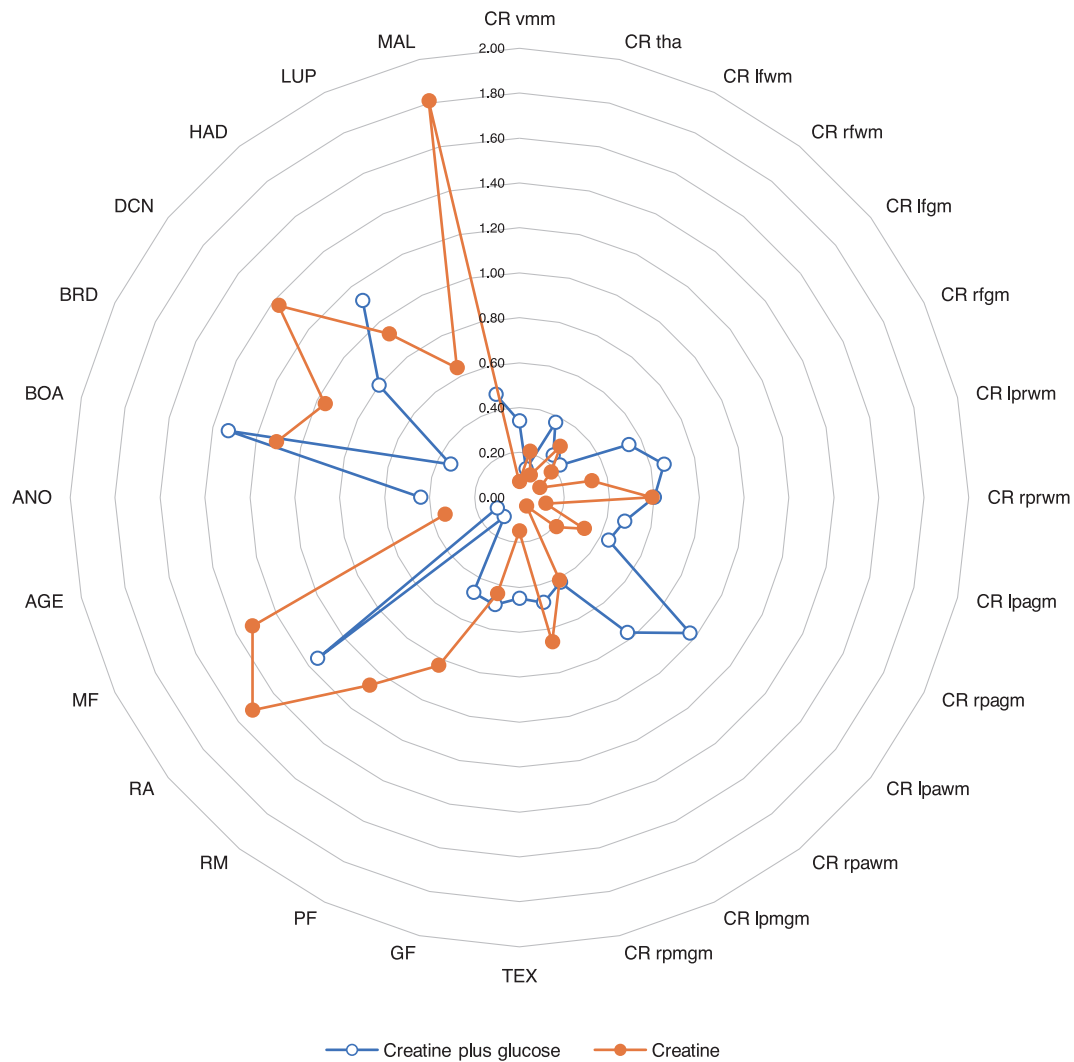


Fig. 1. The Cohen's effects sizes (d) in experimental group 1 (creatine) and experimental group 2 (creatine plus glucose) for all outcomes evaluated in the trial. The effects are classified as small ($d=0.2$), medium ($d=0.5$), and large ($d \geq 0.8$). Missing values are due to a statistical limitation in calculating effect sizes when pre or post values equal zero. CR vmm, total creatine levels (CR) at vastus medialis muscle; CR tha, thalamus; CR lfwm, left frontal white matter; CR rfwm, right frontal white matter; CR lfgm, left frontal grey matter; CR rfgm, right frontal grey matter; CR lprwm, left precentral white matter; CR rprwm, right precentral white matter; CR lpagm, left paracentral grey matter; CR rpagm, right paracentral grey matter; CR lpawm, left parietal white matter; CR rpawm, right parietal white matter; CR lpmgm, left parietal mesial grey matter; CR rpmgm, right parietal mesial grey matter; TEX, time to exhaustion; GF, general fatigue; PF, physical fatigue; RM, reduced motivation; RA, reduced activity; MF, mental fatigue; AGE, ageusia; ANO, anosmia; BOA, body aches; BRD, breathing difficulties; DCN, difficulties concentrating; HAD, headache; LUP, lung pain; and MAL, malaise.

creatine-glucose for a magnitude of rise in total creatine levels at 8 wk follow-up, it appears that creatine-glucose mixture facilitated significant elevation in tissue creatine for a majority of locations analyzed in our study (eight out of fourteen locations, 57.1%) as compared to creatine only (three out of fourteen locations, 21.4%). Similar findings were found for patient-reported outcomes, where the addition of glucose to creatine significantly alleviated long COVID features in five out of eight symptoms comparing to creatine itself (three out of eight symptoms), with the effect sizes were considered moderate-to-large for both creatine interventions. Our results are in accordance with a primary trial where more prolonged supplementation (6 mo) with 4 g of

creatine per day improved tissue creatine status and mitigated a majority of symptoms in long COVID population (6). The present trial adds to published literature by employing more robust protocol of creatine intake, demonstrating a possible benefit of adding glucose to the creatine intervention in long COVID. Still, additional longitudinal randomized controlled trials with larger heterogeneous samples and mechanistic approach are needed to corroborate the promising effects of creatine across various cohorts with long COVID, while employing varying supplementation regimens and combination protocols. It would be crucial to assess the precise mechanisms through which creatine operates within distinct brain regions, and whether supplementing cre-

atine directly impacts areas linked to the onset of fatigue and malaise in long COVID. Further neuroimaging investigations are also needed to examine neural activity changes in regions exhibiting elevated creatine levels.

Authorship

VS and SMO, research conception and design; JS, MR, NT, JO and VS, experiments; NT, VS and SMO, statistical analysis of the data; JS, MR, NT, JO, VS, DGC, LR, JB, PA and SMO, interpretation of the data; JS, MR, NT, JO, VS, DGC, LR, JB, PA and SMO, writing of the manuscript.

Disclosure of state of COI

SMO and DGC serve as members of the Scientific Advisory Board on creatine in health and medicine (AlzChem LLC). SMO co-owns patent "Supplements Based on Liquid Creatine" at European Patent Office (WO2019150323 A1). SMO has received research support related to creatine during the past 36 mo from the Serbian Ministry of Education, Science, and Technological Development; Provincial Secretariat for Higher Education and Scientific Research; Alzchem Group AG; ThermoLife International; and Hueston Hennigan LLP. SMO does not own stocks and shares in any organization. DGC has conducted industry sponsored research involving creatine supplementation and received creatine donations for scientific studies and travel support for presentations involving creatine supplementation at scientific conferences. DGC as an expert witness/consultant in legal cases involving creatine supplementation. JS, MR, NT, JO, VS, LR, JB and PA declare no conflicts of interest related to this study.

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